

Lab 13: Direct Answers, Traces, and a Majority Vote

Compare short answers to step traces on arithmetic, then vote, then flag an unfaithful trace.

Work in a **Jupyter** notebook with **Python** (NumPy optional). Do notes **13.1–13.3** first. **No LLM API**. The traces are lists of strings we constructed.

First time you implement this. The lecture notes this week already answered what the method is, why it exists, the architecture, the formula, and the tradeoffs. Read that first-time block before these exercises. This lab is the first time you **compute** it, not the first definition.

```
PROBLEMS = [
    {
        "id": "p1",
        "q": "23 + 19",
        "gold": 42,
        "traces": [
            {"steps": ["20 + 19 = 39", "39 + 3 = 42"], "answer": 42},
            {"steps": ["23 + 19 = 32"], "answer": 32},
            {"steps": ["3 + 9 = 12, write 2 carry 1", "2 + 1 + 1 = 4, so 42"], "answer": 42},
            {"steps": ["23 + 10 = 33", "33 + 9 = 52"], "answer": 42},
        ],
    },
    {
        "id": "p2",
        "q": "6 * 7 + 5",
        "gold": 47,
        "traces": [
            {"steps": ["6 * 7 = 42", "42 + 5 = 47"], "answer": 47},
            {"steps": ["6 * 7 = 42", "42 + 5 = 47"], "answer": 47},
            {"steps": ["6 + 7 = 13", "13 * 5 = 65"], "answer": 65},
            {"steps": ["6 * 7 = 42", "42 + 5 = 48"], "answer": 47},
        ],
    },
]
```

Trace `p1` last row: steps compute 52, boxed answer is 42. Trace `p2` last row: steps compute 48, box is 47. Those are **unfaithful**.

1. Parse and vote

```
import re

def last_int(text):
    nums = [int(x) for x in re.findall(r"-?\d+", text)]
    return nums[-1] if nums else None

def step_result(trace):
    return last_int(" ".join(trace["steps"]))
```

1. For each problem, print every `answer` and `step_result`.
 2. Majority vote on `answer` (break ties by first-seen winner). Compare the vote to `gold`.
 3. Does voting beat the first trace alone on `p1`? On `p2`?
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2. Flag unfaithful traces

A trace is unfaithful if `step_result != answer`. (If steps contain several integers, you are using the last one on purpose.)

1. Print a table: problem id, trace index, faithful?, answer correct?.
 2. Count **lucky wins**: `answer == gold` and unfaithful.
 3. Accuracy of the majority vote versus **faithful-only** majority (drop unfaithful traces before voting). If the faithful set is empty, write `None`.
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3. Write-up

Four sentences: what self-consistency voted for on each problem, how many lucky wins you found, why a correct box is not a faithful chain, and one extra check you would add if the steps were free English (not just integers).